LAS VIEWER

Well Log Visualization & Petrophysics Platform

DATABRICKS APPS

Powered by Databricks Apps • Unity Catalog • Serverless

ENTERPRISE-GRADE WELL LOG VISUALIZATION & AI-POWERED PETROPHYSICS

LAS Viewer combines interactive SVG-based log curve visualization, automated quality control, configurable processing recipes, and Foundation Model API-powered petrophysical advisory into a single platform for upstream well log analysis.

◆ BUSINESS VALUE

- **Accelerate log interpretation** with interactive multi-track SVG display of raw, corrected, and derived curves
- **Improve data quality** with automated QC detecting spikes, gaps, range violations, and washouts
- **Standardize workflows** with configurable processing recipes from corporate standard to high-fidelity reservoir sim
- **Enhance decision-making** with AI-powered petrophysical recommendations for completion strategies
- **Centralize well data** in Lakebase PostgreSQL with formation tops, curve quality, and processing history

◆ CHALLENGES ADDRESSED

- **Scattered log data** — LAS files spread across network drives with no centralized access
- **Manual QC** — hours spent visually scanning curves for anomalies and data gaps
- **Inconsistent processing** — different petrophysicists applying different correction workflows
- **Limited AI integration** — no automated interpretation or completion recommendation capability
- **Poor visualization** — legacy desktop tools with no web-based collaborative viewing

◆ WHO IT'S FOR

🔬 **Petrophysicist**
Interactive log curve display, QC dashboards, and processing recipe management

🛢️ **Drilling Engineer**
Well status overview, formation boundaries, and data quality scores

📊 **Reservoir Engineer**
Derived porosity, water saturation, and clay volume for characterization

🤖 **Completions Engineer**
AI-powered advisor for well completion and development strategies

6	9	6	3
Wells	Log Curves	Dashboard Tabs	Processing Recipes

◆ QUICK START GUIDE

1. **Clone Repository**
   Clone the LAS Viewer repo to your local machine
2. **Build Frontend**
   Run cd frontend && npm install && npm run build
3. **Deploy the App**
   Run databricks apps deploy to launch the platform
4. **Explore Logs**
   App auto-seeds sample well data on first startup

◆ GETTING THE MOST OUT OF LAS VIEWER

📊 **Browse Log Curves**
Use the interactive SVG viewer to display raw, QC-flagged, corrected, and derived curves

🔍 **Run Automated QC**
Check per-curve quality scores for spike detection, gap identification, and washout flagging

⚙️ **Apply Recipes**
Execute corporate standard, fast turnaround, or high-fidelity processing workflows

🤖 **Consult the Advisor**
Get AI-powered petrophysical recommendations for completion strategies

◆ TARGET INDUSTRY

Upstream Oil & Gas — Well logging and petrophysical analysis for exploration, appraisal, and development wells across any basin or formation type.

⚠ **DEMO ONLY** — This application is a demonstration Databricks App. Please refer to the included LICENSE file for full terms of use.

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